THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 3: **A Story of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number 100 written in base 4 is: 1

(a) 1211 **(b)** 1203 **(c)** 0121 **(d)** 1210

2. A Babylonian numeral has 4 group(s) of 60 and 5 units. Its value is: 1

(a) 9 **(b)** 235 **(c)** 245 **(d)** 29

3. A Maya numeral has 14 in the 20s place and 9 in the units place. Its value is: 1

(a) 149 **(b)** 271 **(c)** 289 **(d)** 23

4. Brahmagupta (628 CE) gave rules for operations with: 1

(a) zero and negative numbers **(b)** Roman numerals only
(c) binary numbers **(d)** fractions only

5. Which of the following is NOT a correct Roman numeral? 1

(a) CD **(b)** VL **(c)** XC **(d)** XL

6. The decimal value of 254_6 is: 1

(a) 108 **(b)** 111 **(c)** 104 **(d)** 106

7. The number 95 written in base 6 is: 1

(a) 532 **(b)** 234 **(c)** 240 **(d)** 235

8. A Babylonian numeral has 5 group(s) of 60 and 30 units. Its value is: 1

(a) 60 **(b)** 270 **(c)** 35 **(d)** 330

9. A Maya numeral has 2 in the 20s place and 18 in the units place. Its value is: 1

(a) 20 **(b)** 38 **(c)** 58 **(d)** 22

10. The base of the Maya number system was: 1

(a) 10 **(b)** 60 **(c)** 5 **(d)** 20

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A Babylonian numeral is made of 5 sixties and 12 units, and a Maya numeral has 8 twenties and 14 units. Find their values. 2

12. Convert 145_6 to a decimal number. 2

13. Write 53 in the binary system. 2

14. A Babylonian numeral is made of 4 sixties and 33 units, and a Maya numeral has 9 twenties and 3 units. Find their values. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Convert 250 to base 4 and check your answer by converting back to decimal. 3

16. Find XLIX + CMLI and write the answer in Roman numerals. 3

17. Write 30213_4 in expanded form and find its decimal value. 3

OR

Convert 333 to base 7 and check your answer by converting back to decimal. 3

18. Find CCCLXII + CDXXXVIII and write the answer in Roman numerals. 3

OR

Write 6150_7 in expanded form and find its decimal value. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Add the binary numbers 11011_2 and 1001_2. Verify your answer by converting to decimal. 5

OR

A shopkeeper has weights of 1, 2, 4, 8, 16 and 32 kg. Show how to weigh exactly 44 kg using these weights, each at most once. Explain the idea using binary numbers. 5

20. Add the binary numbers 10110_2 and 10010_2. Verify your answer by converting to decimal. 5

OR

A shopkeeper has weights of 1, 2, 4, 8, 16 and 32 kg. Show how to weigh exactly 51 kg using these weights, each at most once. Explain the idea using binary numbers. 5

END OF QUESTION PAPER — Total: 40 Marks

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